

Lecture 14: Dynamic Memory Allocation

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Today's Class

- Dynamic memory allocations
- Fragmentation
- Allocation using implicit list
- Allocation using buddy allocator

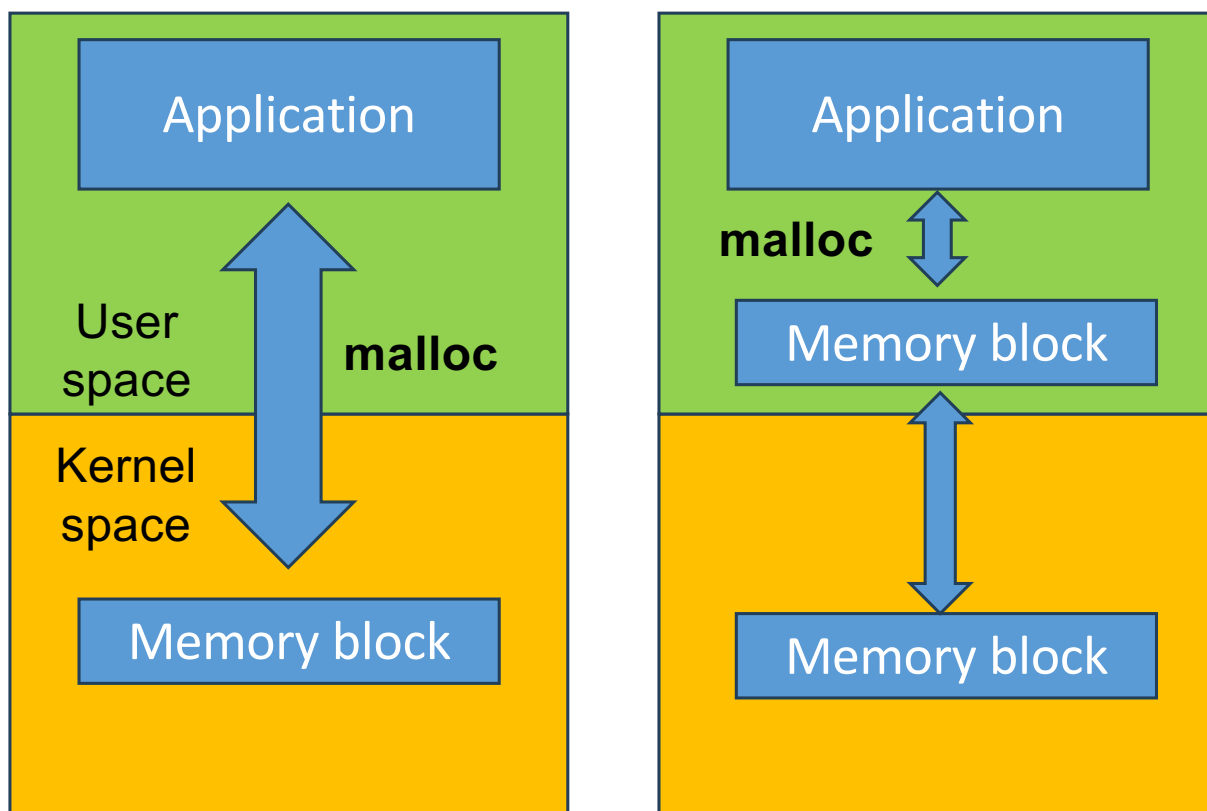
Memory Allocations

- Static size, static allocation
 - Global and static variables
 - Linker allocates final addresses
 - Executable stores these allocated addresses
- Static size, dynamic allocation
 - Local variables
 - Compiler directs stack allocation
- Dynamic size, dynamic allocation
 - Programmer controlled
 - **Allocated in the heap – how?**

Dynamic Memory Allocation

- Explicit vs. implicit memory allocator
 - Explicit: application allocates and *fre*es space
 - e.g., malloc and free in C
 - Implicit: application allocates, but does not free space
 - e.g., garbage collection in Java or Python
- Allocation
 - In both cases the memory allocator provides an abstraction of memory as a set of blocks
 - Gives out free memory blocks to application

Which is Better and Why?



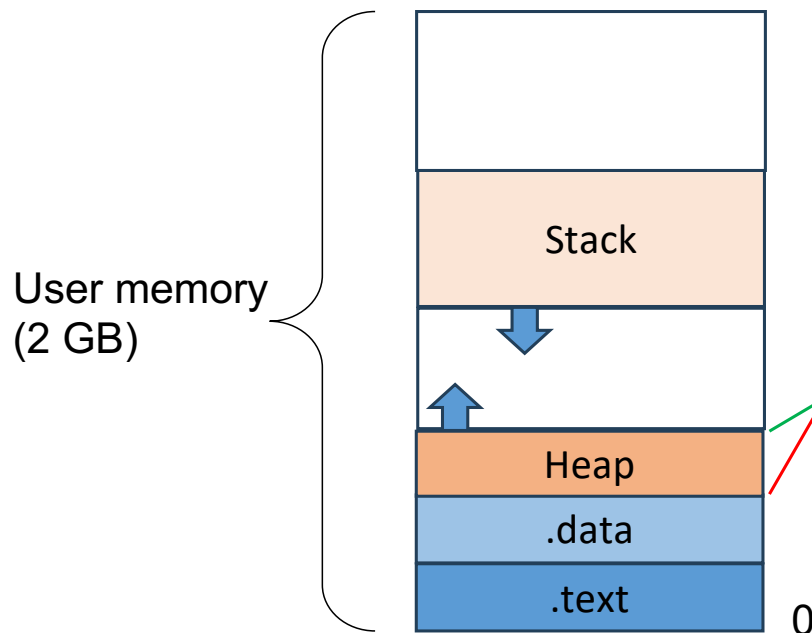
● Case-1

- Switch from user space to kernel space at every call to malloc
- Huge overheads in case of several calls to malloc

● Case-2

- A big chunk of memory is allocated by the OS and mapped into process's address space
- Calls to malloc then returns memory allocated inside the process address space
- Switch from user space to kernel space is significantly reduced

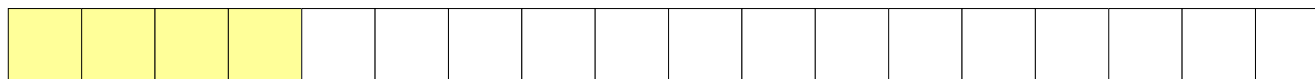
Process Memory Image



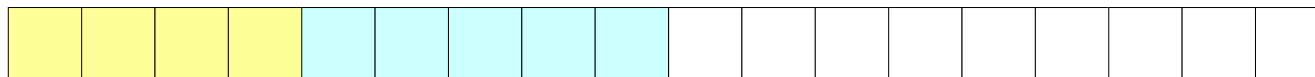
- `void *sbrk(intptr_t nBytes)`
 - Traditionally used by the memory allocators to request or release heap space from the OS
 - **brk** sets the program break to a specific address defining the end of the heap
 - **sbrk** adjusts program break by `nBytes`
 - Positive `nBytes` implies growing heap
 - Negative `nBytes` implies growing heap
 - Modern allocators often use `mmap`
 - *If you are really interested in understanding allocators then go through jemalloc*
 - <https://jemalloc.net/>

Dynamic Memory Allocation Examples

```
p1 = malloc(4*sizeof(int))
```



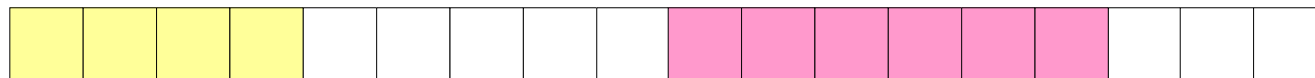
```
p2 = malloc(5*sizeof(int))
```



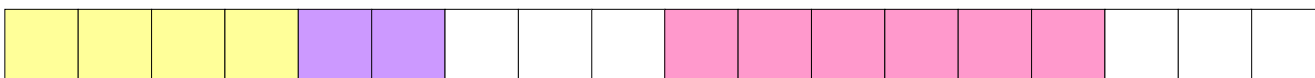
```
p3 = malloc(6*sizeof(int))
```



```
free(p2)
```



```
p4 = malloc(2*sizeof(int))
```



Goals of Good malloc/free

- Good performance for malloc and free
 - Ideally should take constant time (not always possible)
 - Should certainly not take linear time in the number of blocks
- Good space utilization
 - User allocated structures should use most of the heap
 - Want to minimize “**fragmentation**”

Today's Class

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Fragmentation

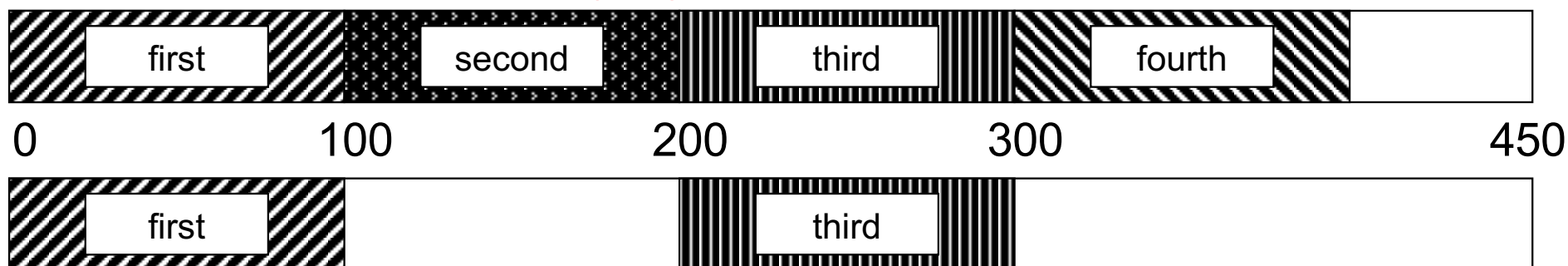
- Segments of memory can become unusable due to the result of allocation scheme
- Two types of fragmentation
 - External fragmentation
 - Memory remains unallocated
 - Variable allocation sizes
 - Internal fragmentation
 - Memory is allocated but unused
 - Fixed allocation sizes

External Fragmentation

- Imagine calling a series of *malloc* and *free*

```
char* first = malloc(100);
char* second = malloc(100);
char* third = malloc(100);
char* fourth = malloc(100);
free(second);
free(fourth);
char* problem = malloc(200);
```

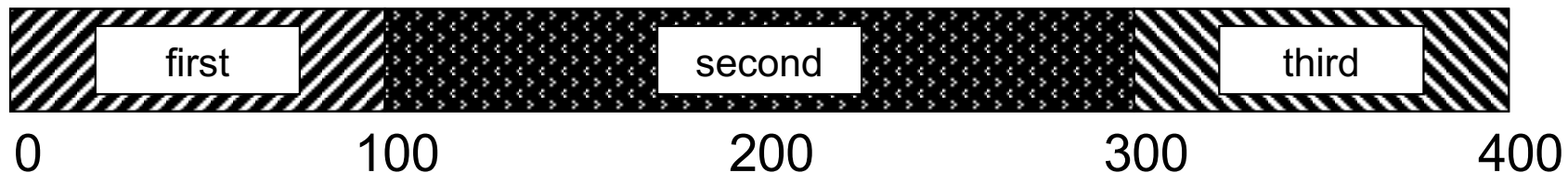
- 250 free bytes of memory, only 150 contiguous
 - unable to satisfy final malloc request



Internal Fragmentation

- Imagine calling a series of *malloc*
 - Assume allocation unit is 100 bytes

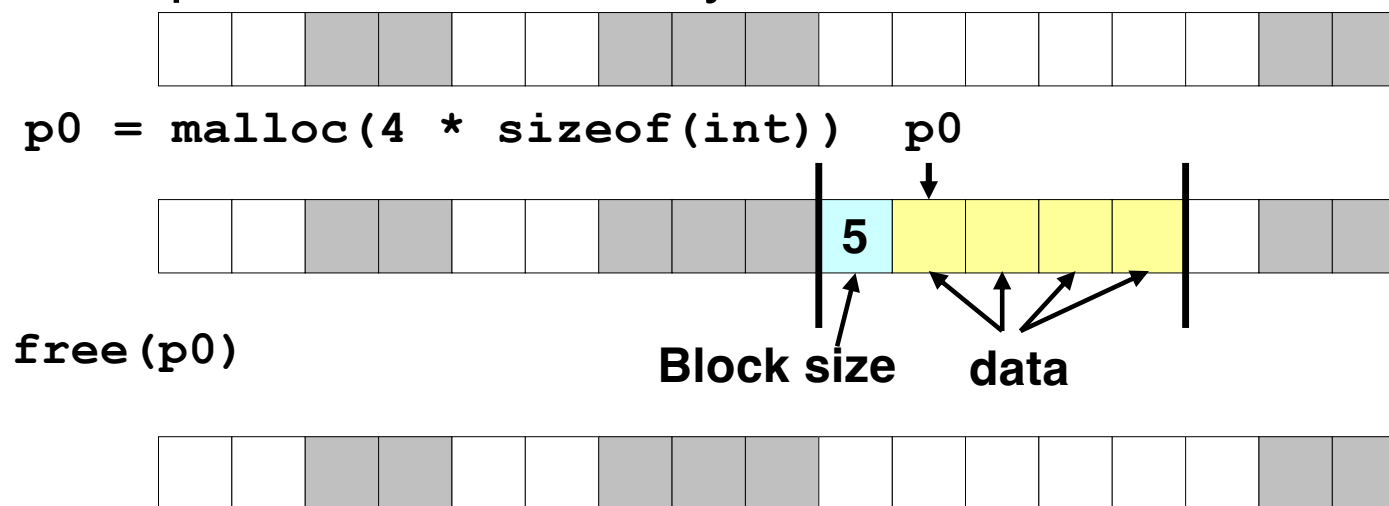
```
char* first = malloc(90);  
char* second = malloc(120);  
char* third = malloc(10);  
char* problem = malloc(50);
```
- All of memory has been allocated but only a fraction of it is used (220 bytes)
 - Unable to handle final memory request



Knowing How Much to Free

● Standard method

- Keep the length of a block in the memory slot preceding the block
 - This slot is often called the header field or header
- Requires a slot for every allocated block

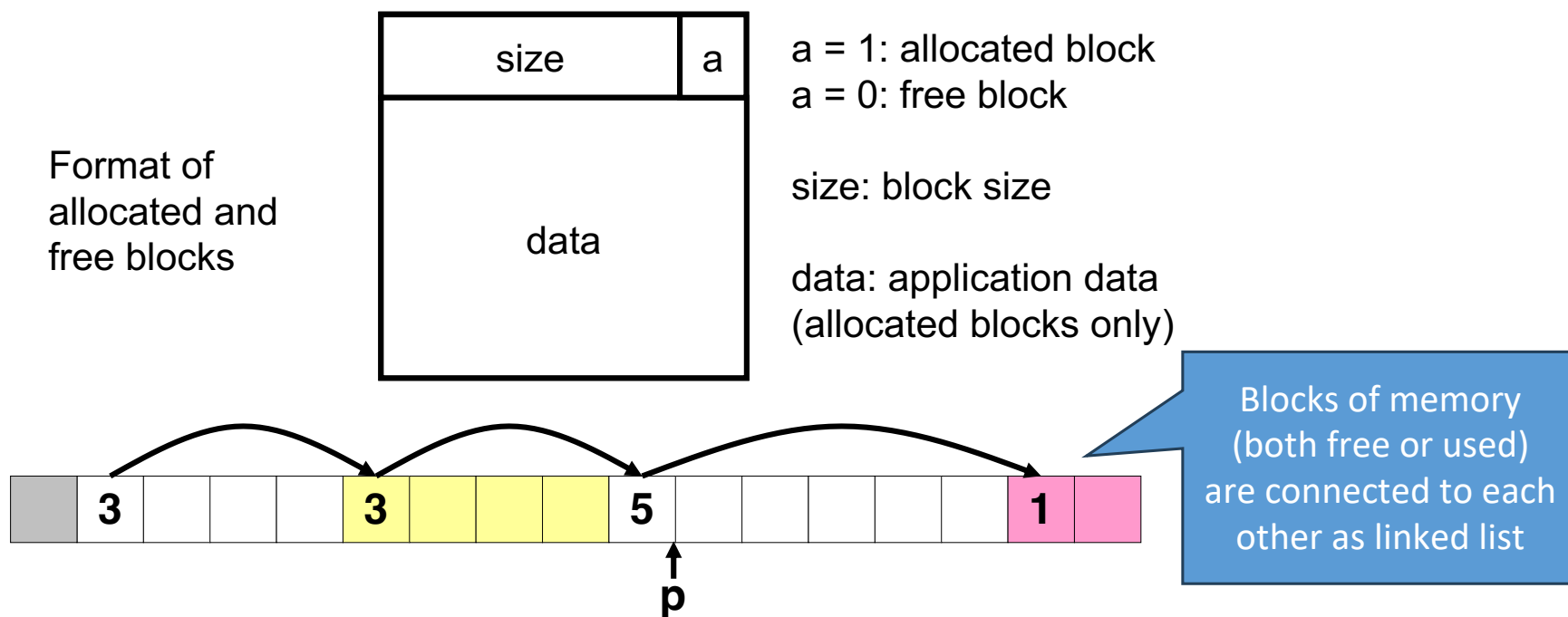


Today's Class

- Dynamic memory allocations
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Implicit List: Keeping Track of Memory Blocks

- Need to identify whether each block is free or allocated
 - Block header at the start of each block of memory
 - Header is a structure with size and status



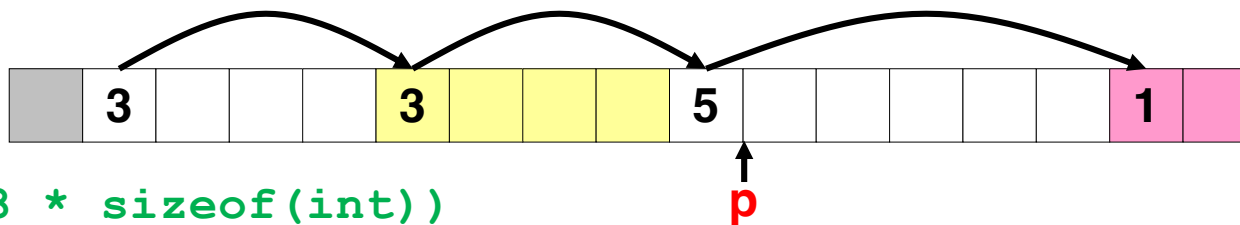
Implicit List: Finding a Free Block

- **First fit:**
 - Search list from beginning, choose first free block that fits
 - May miss a block with closest size that fits
 - Can take linear time in total number of blocks (allocated/free)
- **Next fit:**
 - Like first-fit, but search list from end of previous search
 - Research suggests that fragmentation is worse
 - May miss a block with closest size that fits
- **Best fit:**
 - Choose the free block with the closest size that fits
 - Requires complete search of the list
 - Keeps fragments small – usually helps fragmentation
 - Will typically run slower than first-fit

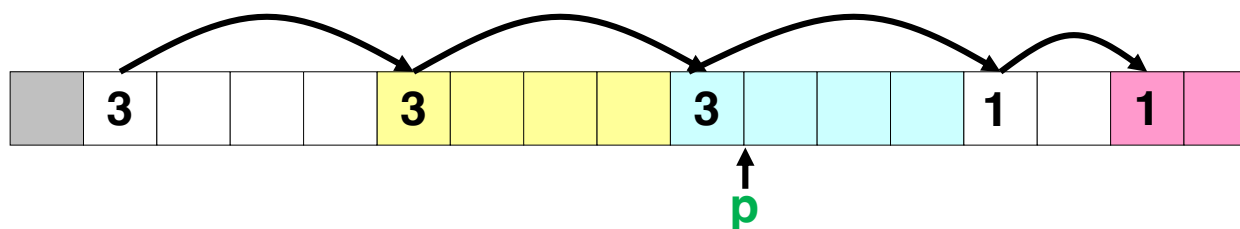
Implicit List: Allocating in Free Block

- Allocating in a free block
 - Since allocated space might be smaller than free space, we might want to split the block (Splitting)

```
p = malloc(3 * sizeof(int))
```

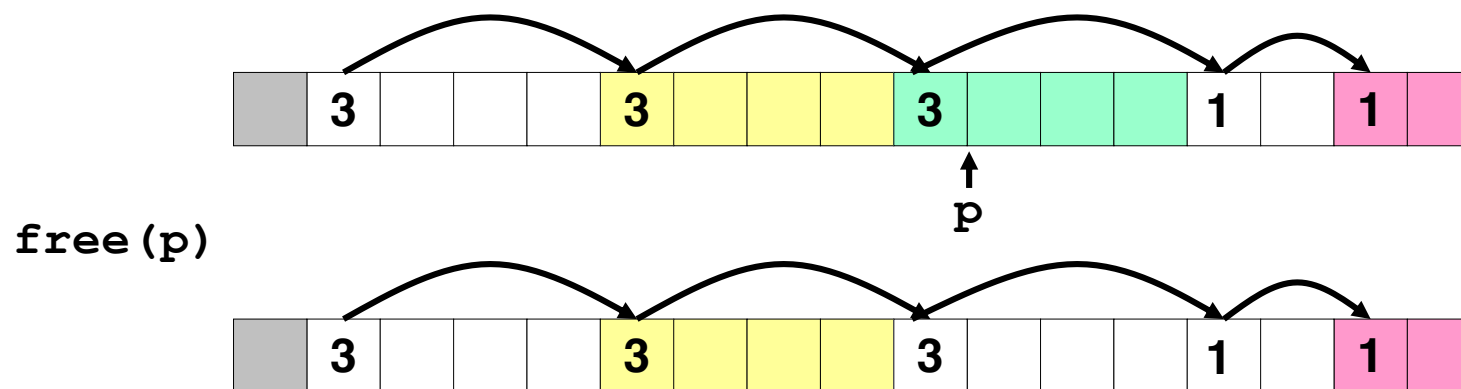


```
p = malloc(3 * sizeof(int))
```



Implicit List: Freeing a Block

- Simplest implementation:
 - Only need to clear allocated flag
 - But can lead to “false fragmentation”

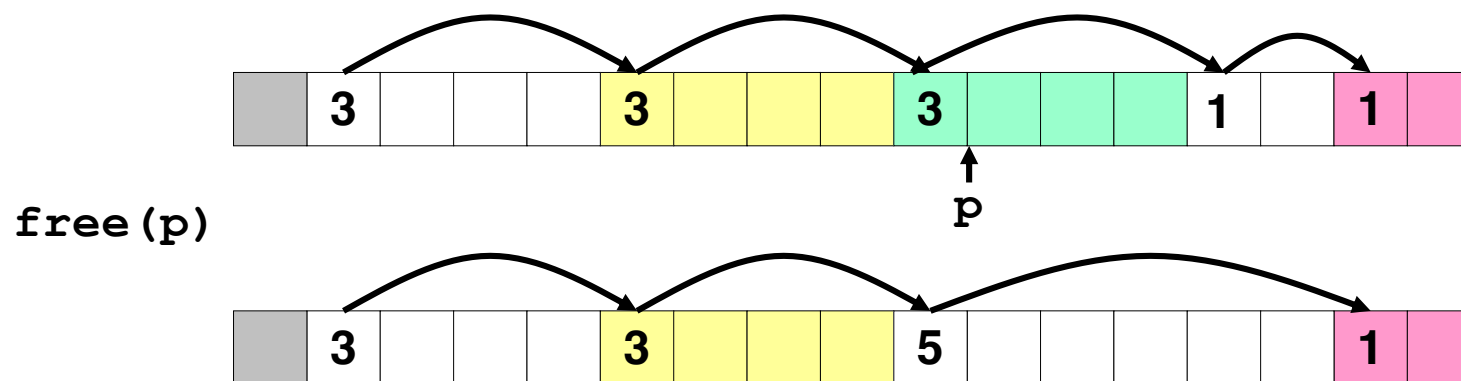


`malloc(4 * sizeof(int))` **Oops!**

- There is enough free space, but the allocator won't be able to find it!

Implicit List: Coalescing

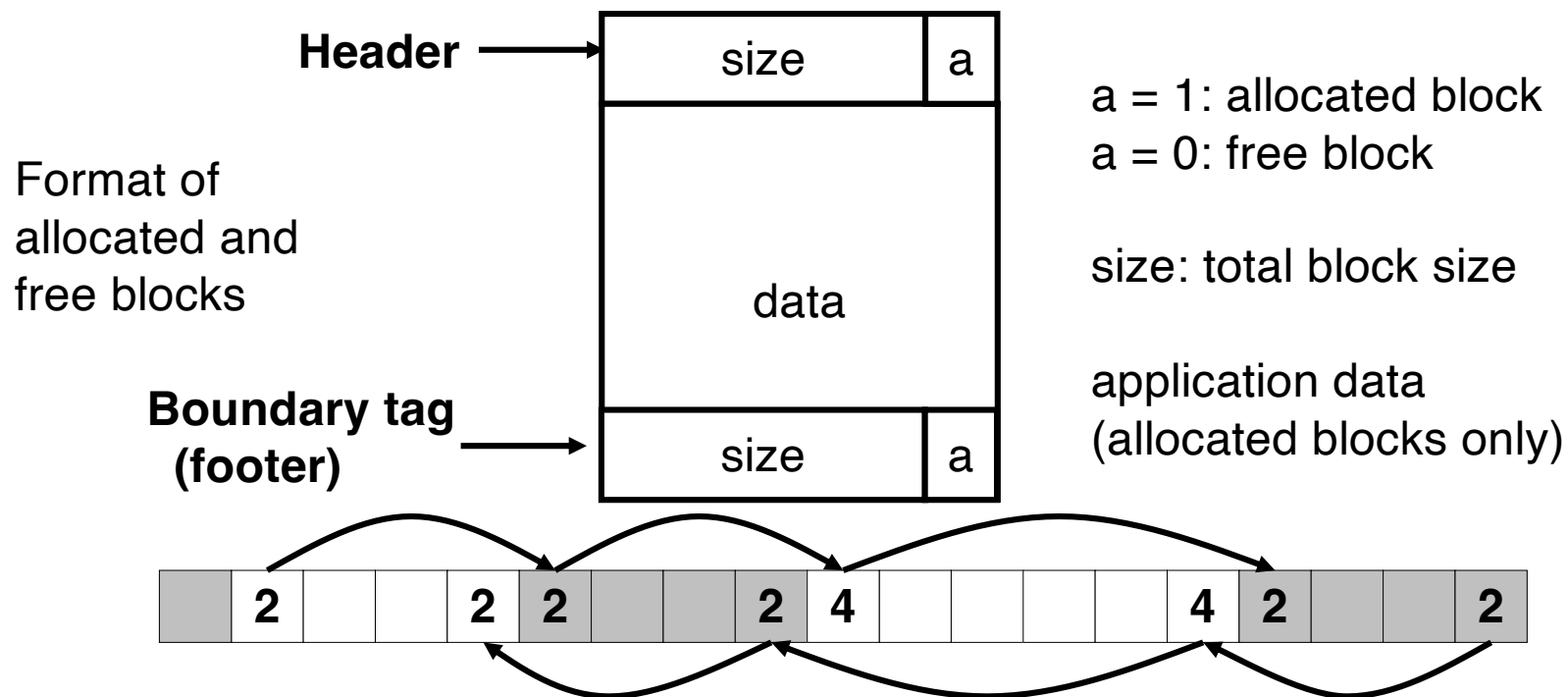
- Join (coalesce) with next and/or previous block **if free**
 - Coalescing with **next** block



- But how do we coalesce with previous block?

Implicit List: Bidirectional Coalescing

- Boundary tags [Knuth73]
 - Replicate header at end of block
 - Allows us to traverse the “list” backwards, but requires extra space



Today's Class

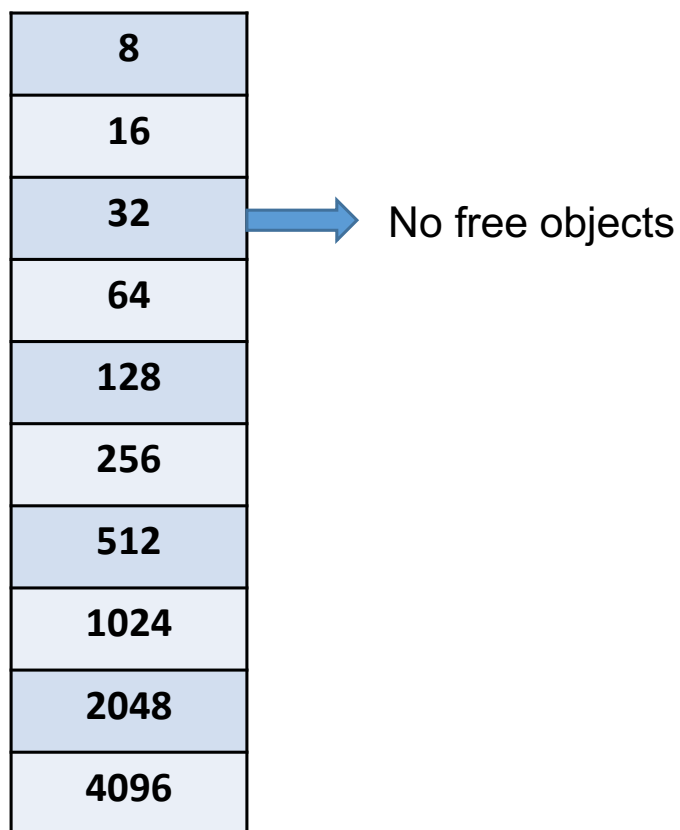
- Dynamic memory allocations
- Fragmentation
- Allocation using implicit list
- Allocation using buddy allocator

Buddy Allocator

8
16
32
64
128
256
512
1024
2048
4096

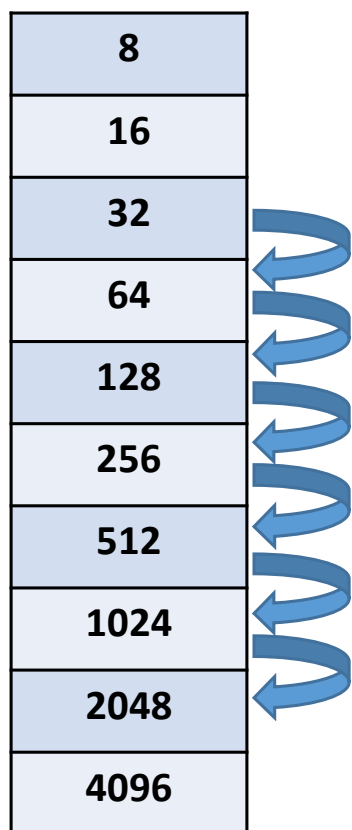
- Implicit list
 - Memory usage depends on placement policy
 - First fit, next fit, or best fit
 - Prone to fragmentation
- Buddy allocator
 - It maintains free buckets of memory objects of different allocation sizes (2^k)
 - Initially all buckets are empty
 - Faster allocation
 - Reduces fragmentation

Buddy Allocator



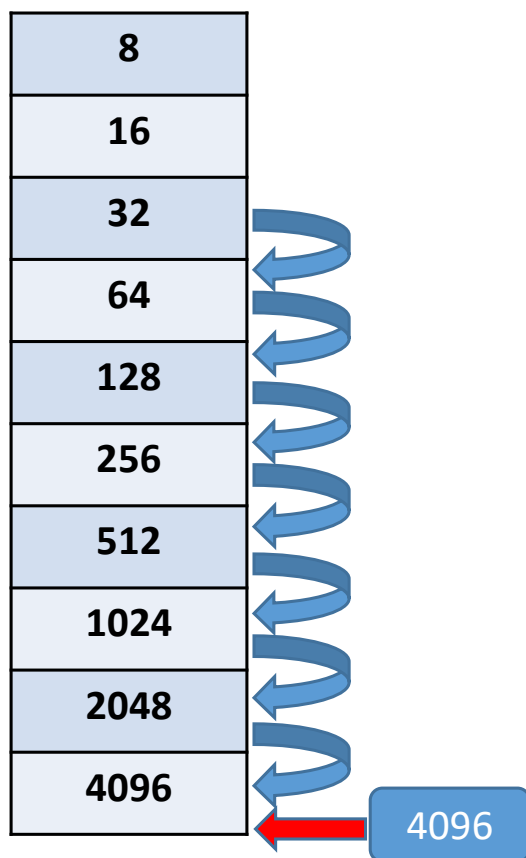
- `A = malloc(24)`
 - Rounded size = 32
 - Little bit of fragmentation, but let's ignore that
 - No free objects in that bucket

Buddy Allocator



- Allocator recursively calls itself to allocate an object that is twice the size of the bucket size
- Call would eventually lead to bucket 4096

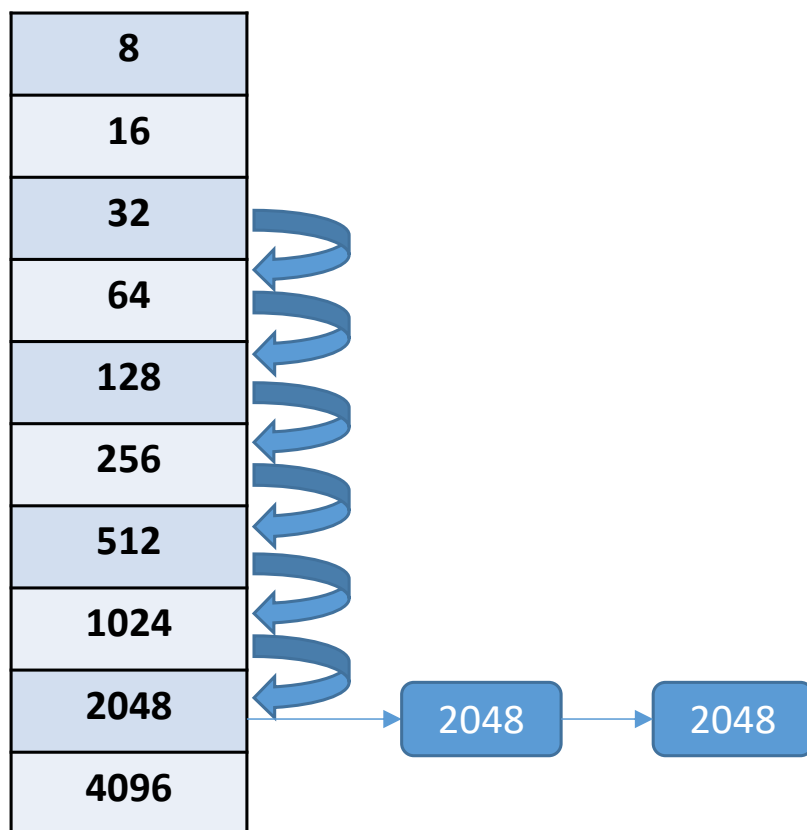
Buddy Allocator



No free objects.
Hence, allocate
4Kb from RAM

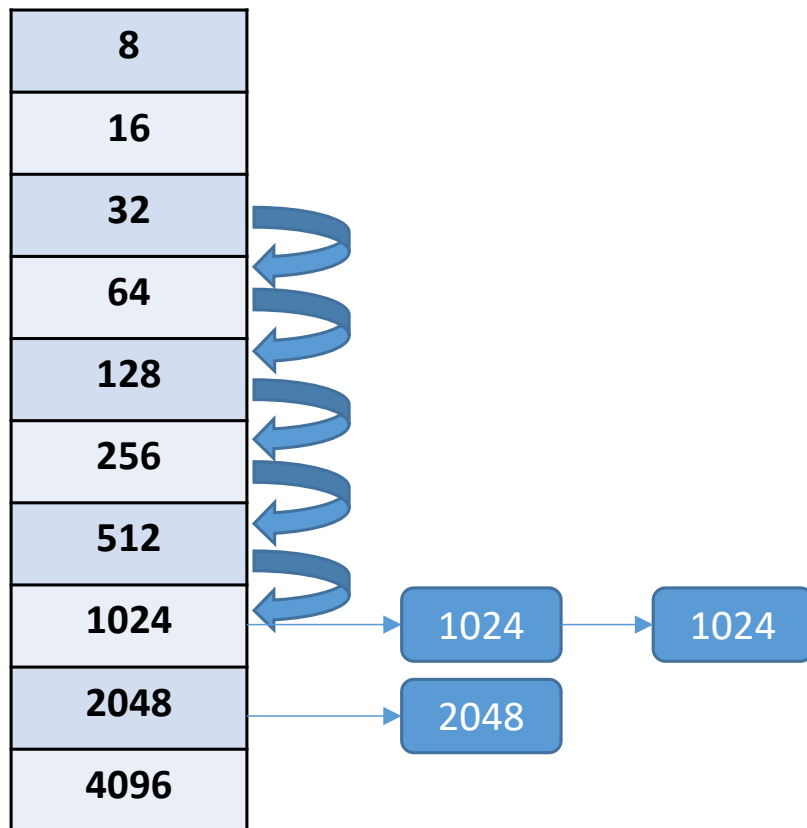
- There is no free object in the bucket size 4096
 - Request OS to allocate a chunk of size 4096 bytes from the RAM and return to its caller
 - System call
 - 4096 bytes is also called as page size in Linux (will be covered in Lecture 17)

Buddy Allocator



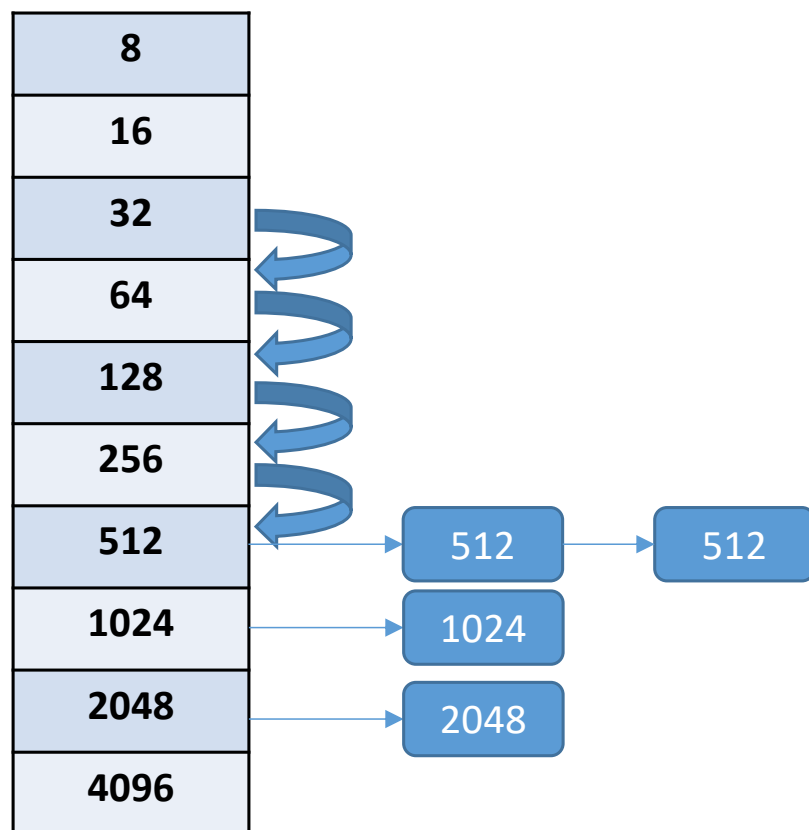
- Split the 4096 object into two halves (two 2048 objects), add into the parent bucket
 - Bucket 2048

Buddy Allocator



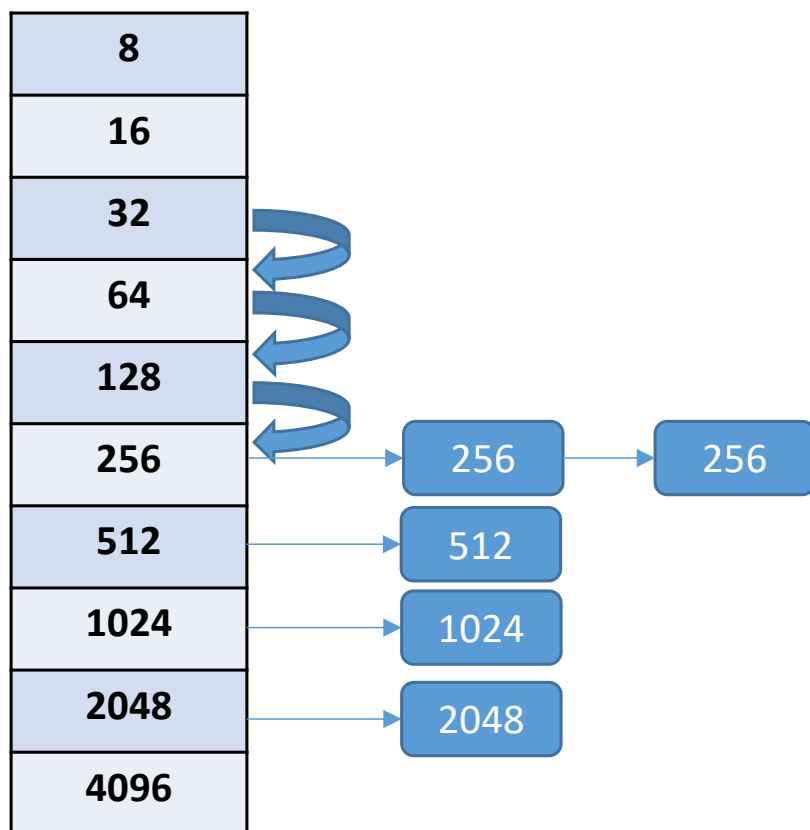
- One of the 2048 object is promoted to the bucket 1024 where it is split into two objects of size 1024 each

Buddy Allocator



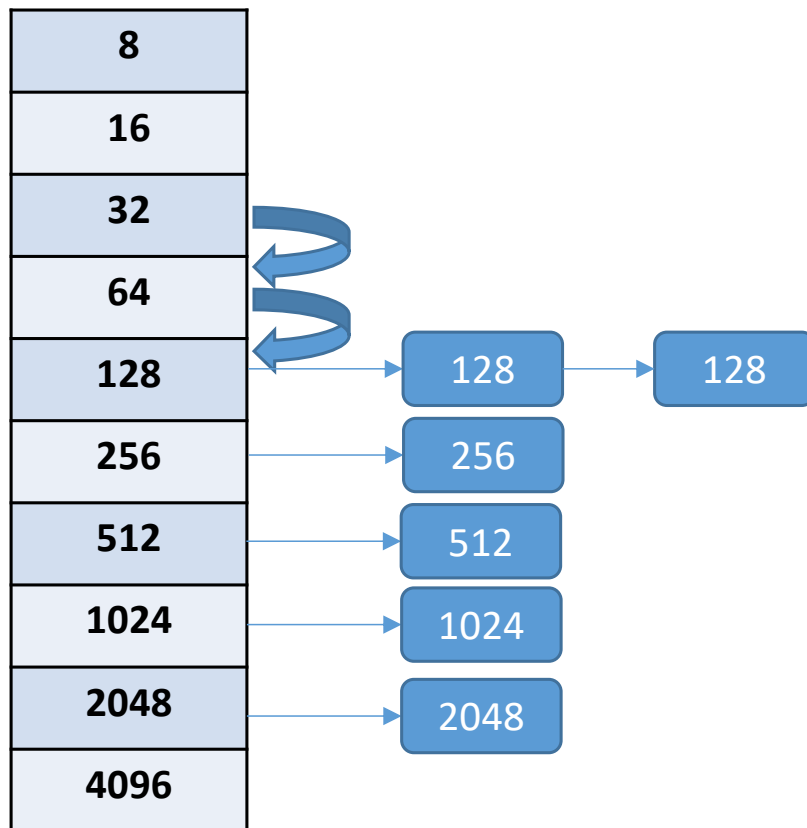
- One of the 1024 object is promoted to the bucket 512 where it is split into two objects of size 512 each

Buddy Allocator



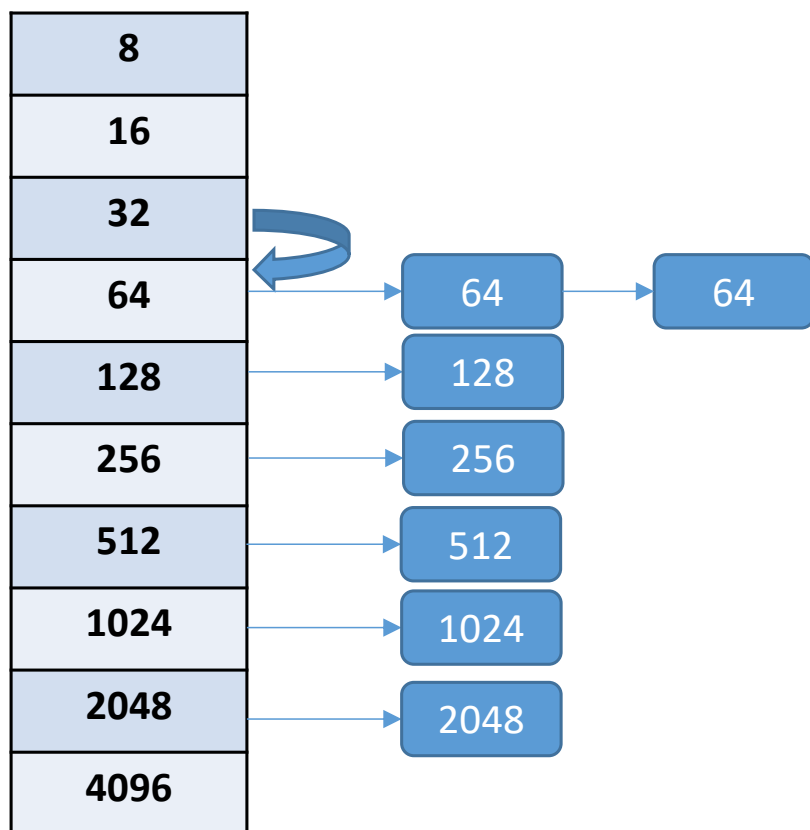
- One of the 512 object is promoted to the bucket 256 where it is split into two objects of size 256 each

Buddy Allocator



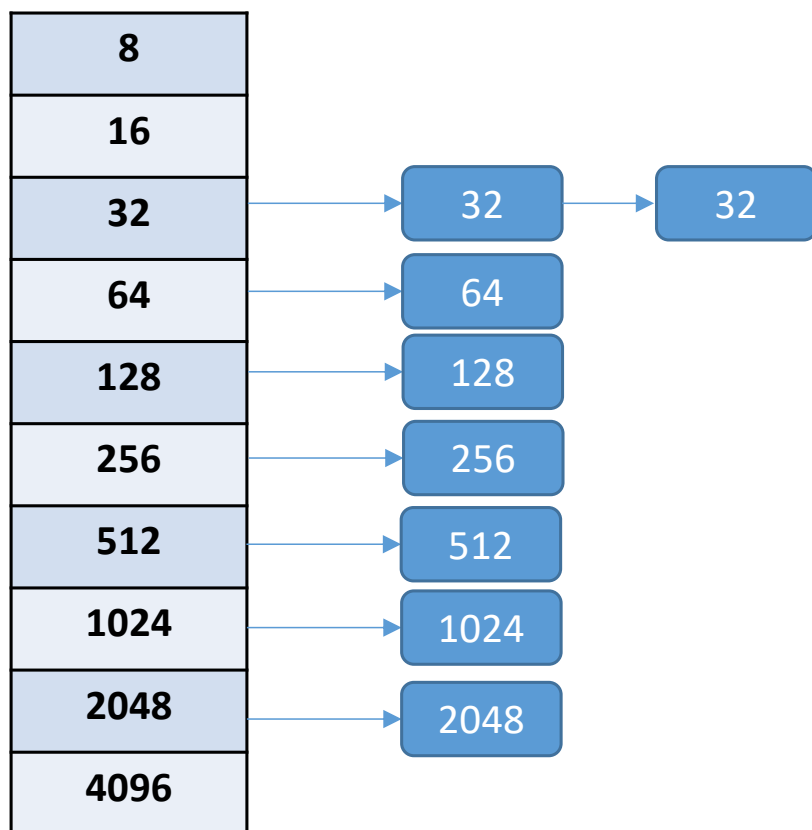
- One of the 256 object is promoted to the bucket 128 where it is split into two objects of size 128 each

Buddy Allocator



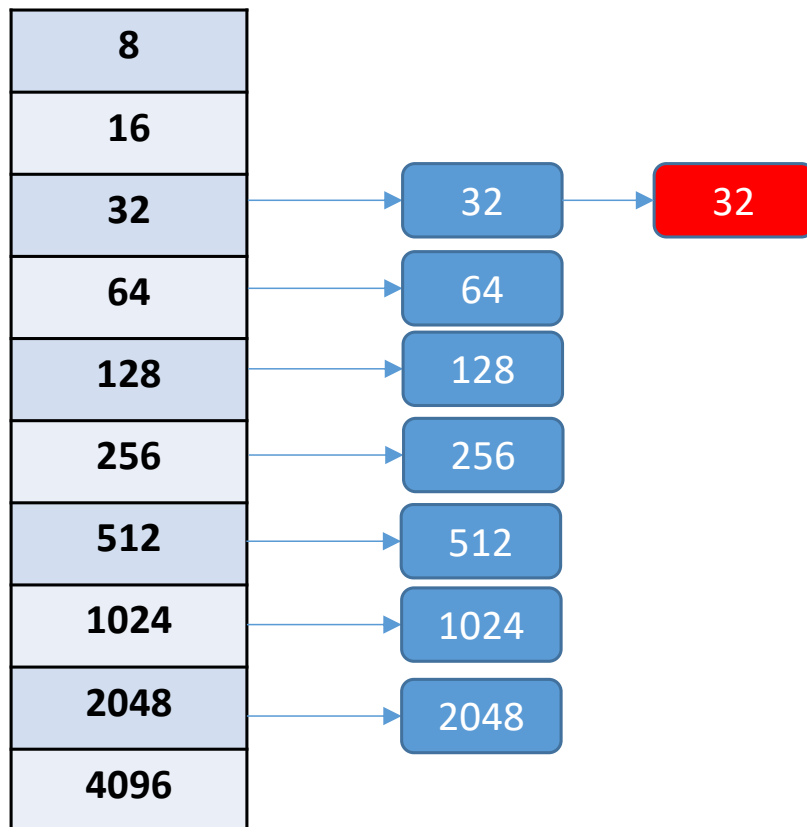
- One of the 128 object is promoted to the bucket 64 where it is split into two objects of size 64 each

Buddy Allocator



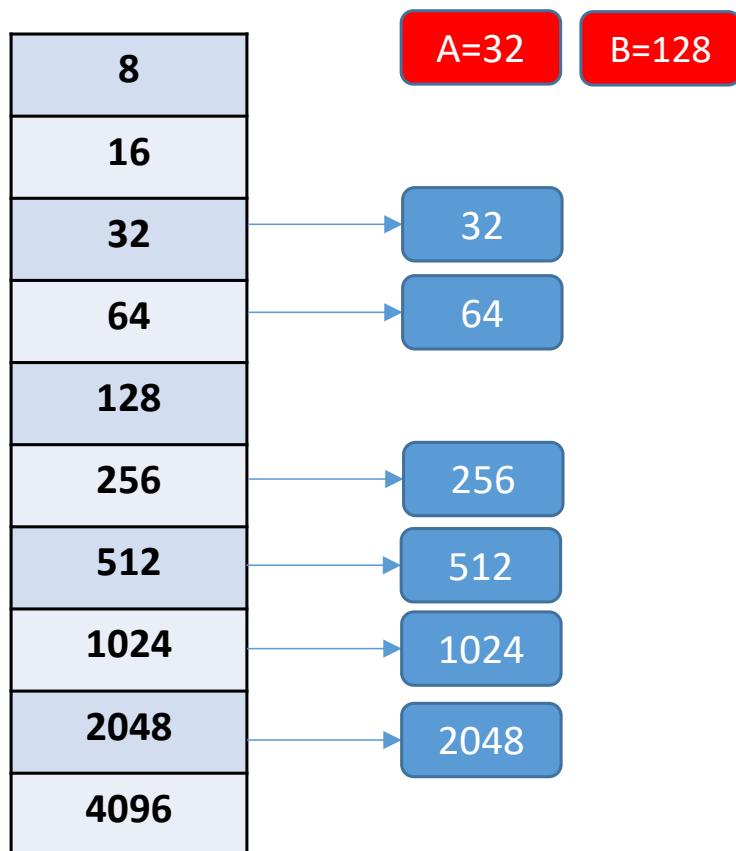
- One of the 64 object is promoted to the bucket 32 where it is split into two objects of size 32 each

Buddy Allocator



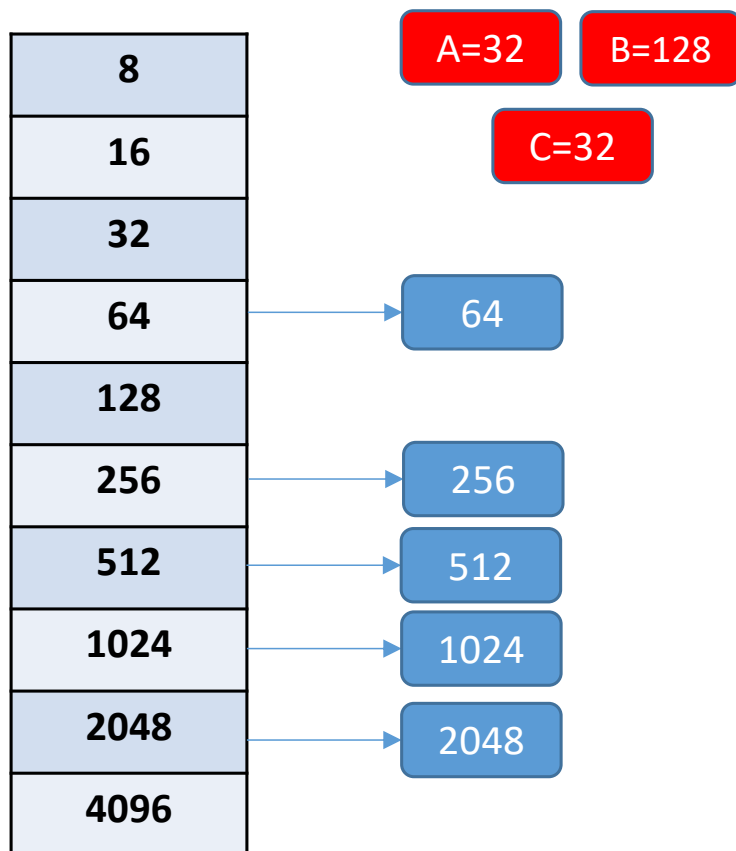
- One of the 32-byte object in bucket 32 will now be returned to the `malloc(32)` call

Buddy Allocator



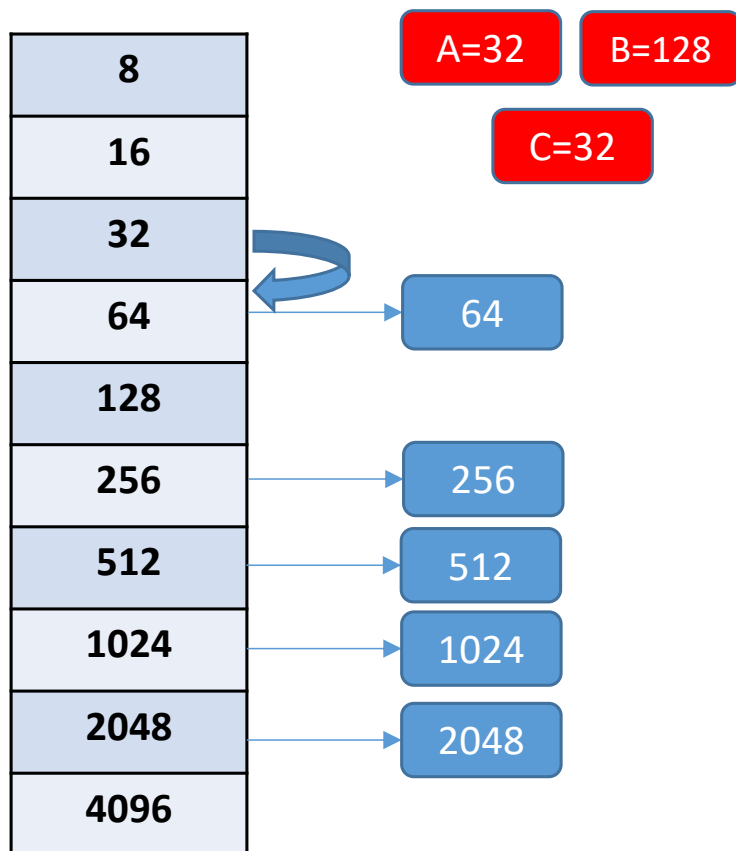
- **B = malloc(128)**
 - One free object already in that bucket 128
 - Return that free object from the bucket to the malloc(128)
 - Now no more free object in bucket 128

Buddy Allocator



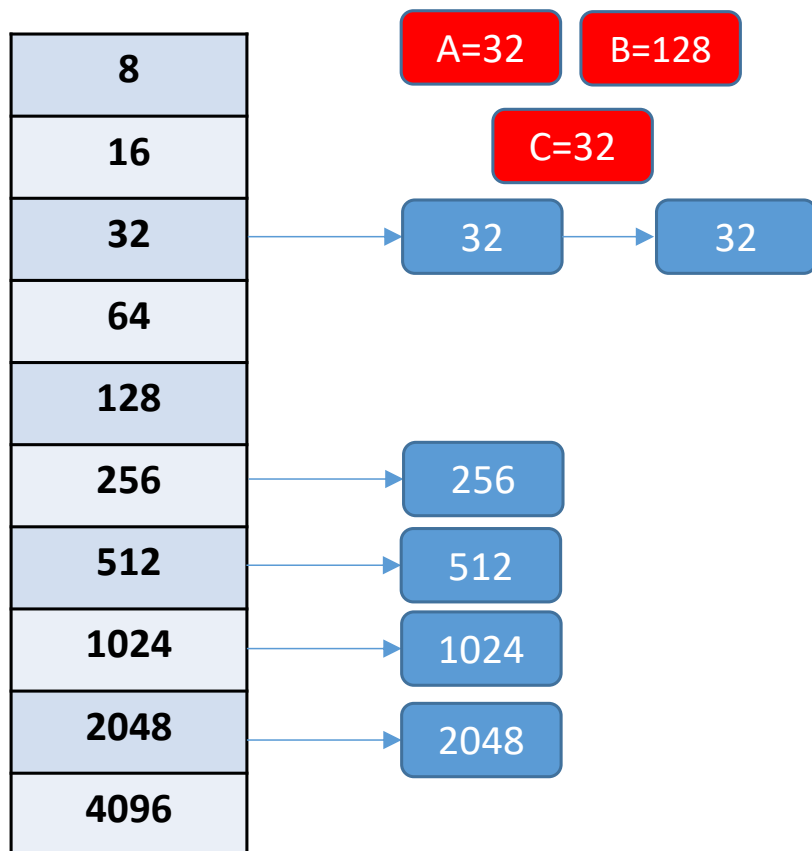
- `C = malloc(32)`
 - One free object already in that bucket
 - Return that free object from the bucket to the `malloc(32)`
 - Now no more free object in bucket 32

Buddy Allocator



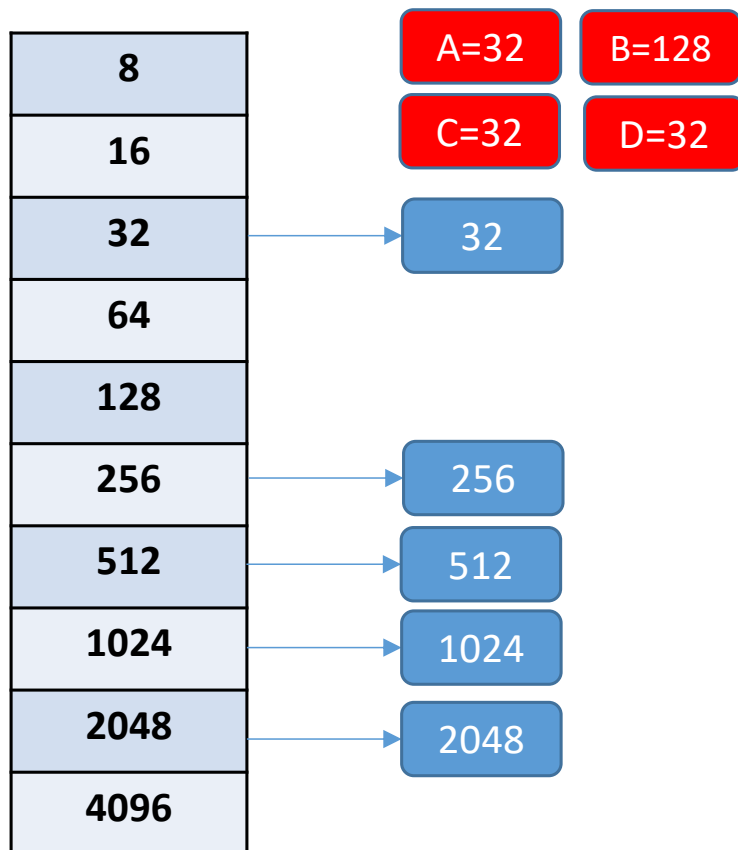
- $D = \text{malloc}(32)$
 - No more free object exists in the bucket 32
 - Fetch one object from the buddy bucket 64

Buddy Allocator



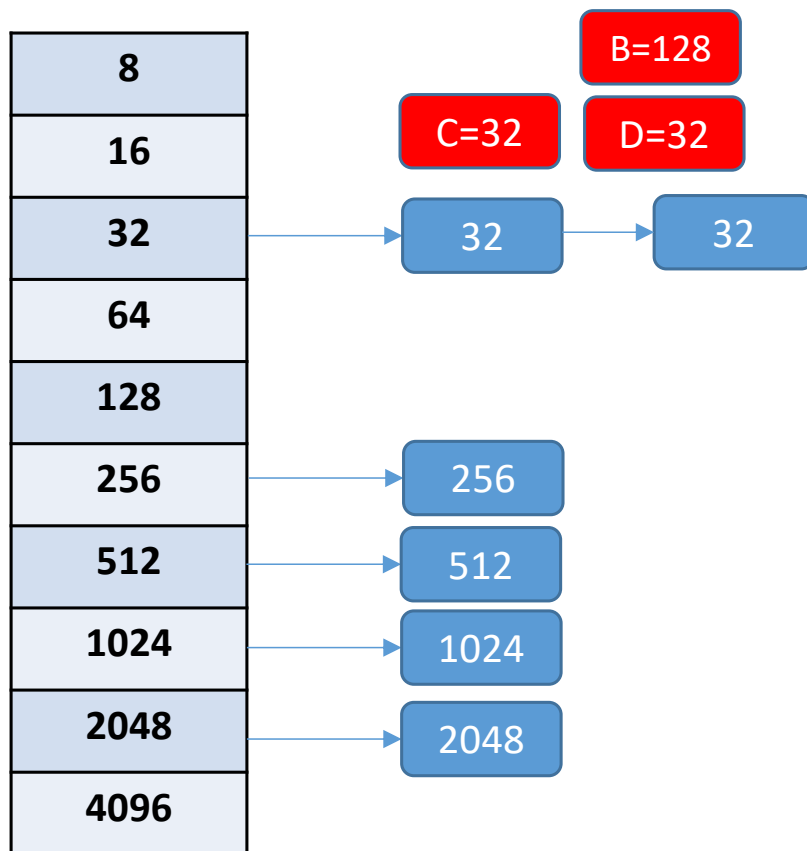
- Object of 64 bytes received from buddy bucket 64 will be split into two 32 byte free objects and added into the bucket 32

Buddy Allocator



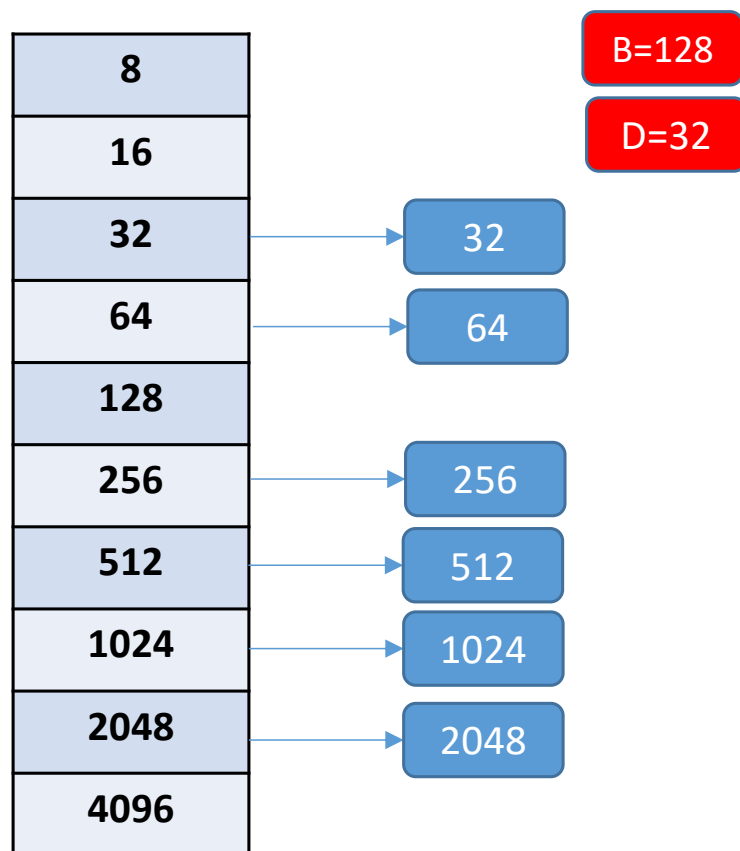
- One free object from bucket 32 returned to malloc(32) call

Buddy Allocator



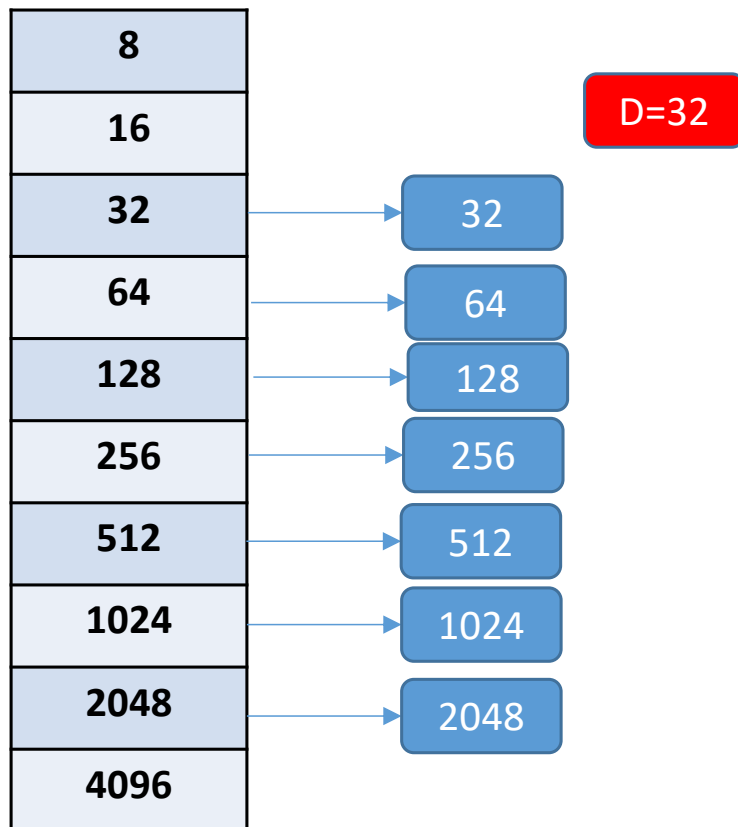
- **free(A)**
 - Append to the free object list in bucket 32

Buddy Allocator



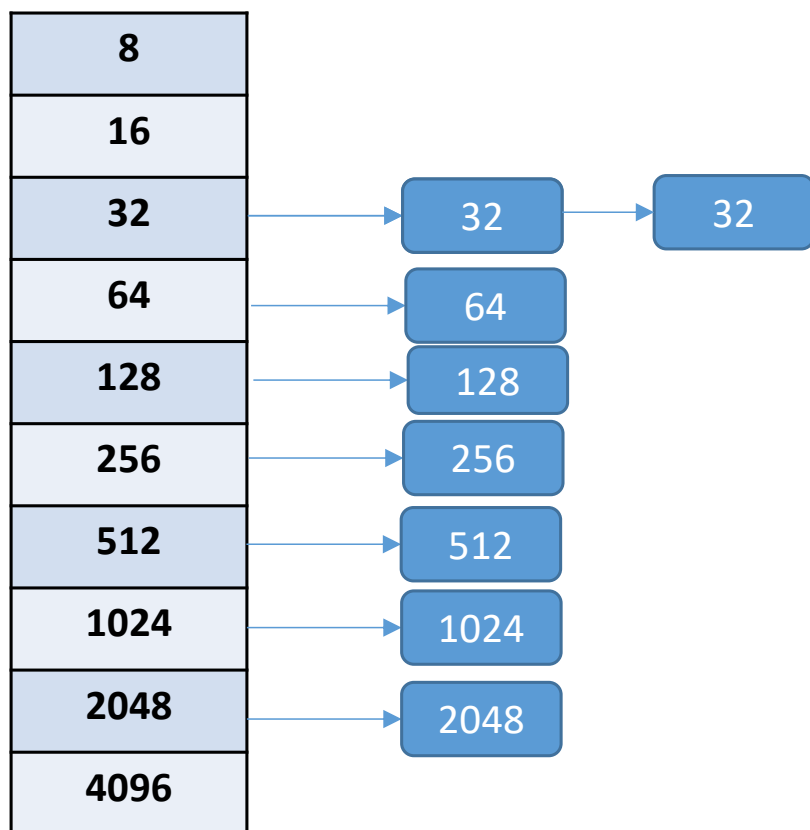
- **free(C)**
 - One more memory object appended to the free list in bucket 32 which results in a total of 3 free objects in that bucket
 - Two of the free objects are coalesced (merged) and promoted to bucket 64
- If merging and promotion doesn't happen then it will result in unnecessary calls to OS for 4Kb objects
 - System call invocation, thereby leading to overheads!

Buddy Allocator



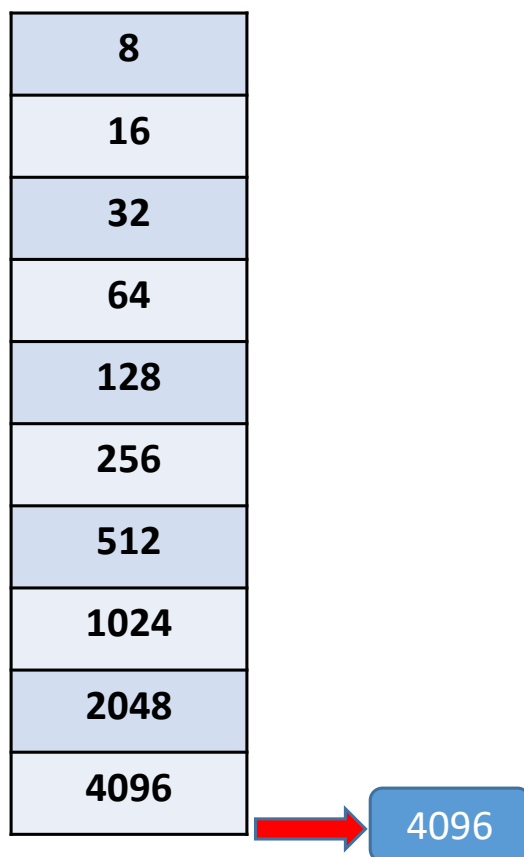
- `free(B)`
 - Append to the bucket 128

Buddy Allocator



- `free(D)`
 - Append the memory object back to the bucket (32)
- Finally, during process termination, all free objects are coalesced back into 4096 bucket and returned to the OS

Buddy Allocator



- All free objects are coalesced back into 4096 bucket and returned to the OS at process termination
 - What would happen if memory not returned to OS?
- How to handle allocations greater than 4096 bytes?
 - The size is rounded up to the nearest multiples of 4096 and allocated directly from OS

Next Lecture

- Virtual memory